

L01 LIGHT EXPOSURE | P

Intent : Provide indoor light exposure through daylight and electric light strategies.

Summary : This WELL feature requires projects to provide appropriate light exposure in indoor environments through lighting strategies.

Issue : Mammals function on an approximately 24-hour cycle, following what are referred to as circadian rhythms.¹ Light is the main driver of the circadian system, which controls body processes, such as digestion, the release of certain hormones, body temperature and sleep. Humans evolved their sleep schedules to be dependent on the natural day- night cycle. Electric light has only been widely used as a source of light for less than 200 years.² The increase in the use of electric lighting has led to indoor environments relying on electric light over daylight. However, reduced exposure to light has been linked to the onset of depression and impairment of cognitive function in individuals.^{3,4} Irregular sleep-wake cycles have been linked to poorer academic performance in college students.⁵ Studies have shown that light exposure during daytime has been linked to increases in productivity and sleep quality, as well as reduction in symptoms of depression in individuals.⁶⁻⁸

Solutions : Providing indoor access to adequate light can positively influence the productivity and mood of individuals, while supporting the alignment of their circadian rhythms with the natural day-night cycle.⁹⁻¹² Access to appropriate levels of light in indoor environments can be achieved through building design, façade design, space layout and lighting design. Windows, atriums and skylights are design features that can be utilized to increase daylight in a space. The interior layout of the space also has an impact on the daylight exposure received by users. For example, conference rooms can be added to the center of the floor plate, so that workstations can be situated near windows and have daylight exposure. Lighting strategies using electric lighting can be utilized to achieve required light exposure, when appropriate daylight exposure is not available.

PART 1 PROVIDE INDOOR LIGHT

For All Spaces except Dwelling Units & Guest Rooms:

Option 1: Daylight simulation

The project demonstrates, through computer simulations, that one of the following conditions are achieved:

- a. The total floor area of regularly occupied spaces achieves one of the following targets:

**Calculations per IES
LM-83¹³**

Calculations per Annex A of CEN 17037

$sDA_{150,50\%} > 30\%$

OR Target illuminance 200 lux is achieved for >30% of floor area throughout 50% of daylit hours of the year

- b. Common space that has unassigned seating for at least 15% of regular occupants at any given time achieves one of the following targets:

**Calculations
per IES LM-
83**

Calculations per Annex A of CEN 17037

$sDA_{300,50\%} > 75\%$

OR Target illuminance 300 lux is achieved for >30% of floor area and average illuminance 100 lux is achieved for >95% of floor area throughout 50% of daylit hours of the year

OR

Option 2: Interior layout

One of the following requirements is met:

- a. At least 30% of the regularly occupied area is within a 6 m horizontal distance of envelope glazing.
- b. Common spaces have unassigned seating and can accommodate at least 15% of regular occupants at any given time. At least 70% of all seating in the spaces is within a 5 m horizontal distance of envelope glazing.

OR

Option 3: Building design

One of the following requirements is met:

- a. The envelope glazing area is no less than 7% of the regularly occupied floor area.
- b.

The floor plate is no more than 20 m between opposite walls that each have envelope glazing, and there are no opaque obstructions higher than 1.22 m within a 6 m horizontal distance of the envelope glazing.

OR

Option 4: Circadian lighting design

The following requirement is met:

- a. The project meets the threshold of Tier 1 of Feature L03: Circadian Lighting Design.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

For Dwelling Units & Guest Rooms:

Option 1: Daylight simulation

The project demonstrates, through computer simulations, that the following requirement is achieved:

- a. The total floor area of regularly occupied spaces of each dwelling unit achieves one of the following targets:

**Calculations per IES
LM-83¹³**

sDA_{150,50%} > 30%

Calculations per Annex A of CEN 17037

OR Target illuminance 200 lux is achieved for >30% of individual unit area throughout 50% of daylit hours of the year

OR

Option 2: Facade design

The following requirement is met:

- a. The envelope glazing area is no less than 7% of the regularly occupied floor area for each dwelling unit.

OR

Option 3: Circadian lighting design

The following requirement is met:

- a. The project achieves at least one point in Feature L03: Circadian Lighting Design.

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L02 VISUAL LIGHTING DESIGN | P

Intent : Provide visual comfort and enhance visual acuity for all users through electric lighting.

Summary : This WELL feature requires projects to provide appropriate illuminances on work planes for regular users of all age groups, as required for the tasks performed in the space.

Issue : Humans perceive the world through visual cues that are received through images formed on the retina of the eye. The light levels in a space can enhance the user's ability to perform tasks in that space, while contributing to the feeling of spaciousness. The age of the individual is also a factor in the amount of light required for visual acuity. As humans age, the transmission of light through their lenses is reduced. This is due to age-related changes, including increased light absorption by the lenses, smaller pupil size, increased scattering of light due to thicker lenses and yellowing of the lenses.^{1,2} This aging of the eye indicates that an increase in light levels is required to ensure visual acuity.

Solutions : While developing a lighting strategy to accommodate the visual acuity of users, it is critical to take into account the tasks conducted, as well as the age of the users. Projects may refer to published recommendations by lighting associations or authorities on using electric lighting design strategies for light levels required on the work plane. Lighting recommendations published by authorities provide a range of lighting levels for different age groups and tasks.

PART 1 PROVIDE VISUAL ACUITY

For All Spaces except Dwelling Units:

Option 1: Visual lighting design

The following requirements are met:

- a. All indoor and outdoor spaces (including transition areas) comply with the illuminance thresholds specified in one of the following lighting reference guidelines:
 1. IES Lighting Library, Lighting Applications Standards Collection.³
 2. EN 12464-1:2021⁴ or EN 12464-2:2014.⁵
 3. ISO 8995-1:2002(E) (CIE S 008/E:2001).⁶
 4. GB50034-2013.⁷
 5. CIBSE SLL Code for Lighting.⁸
- b. The illuminance thresholds take into consideration the tasks and the age groups of the occupants.

OR

Option 2: Predetermined light levels

The following requirements are met:

- a. More than 50% of the occupants are under the age of 65.
- b. The area of outdoor space within the project boundary is less than 5% of the interior project area.
- c. At least 90% of the interior project area is comprised of the following space types and meets the associated illuminance thresholds:

Room Types	Minimum Illuminance Threshold
Offices	320 lux at task surface. ⁹
Classrooms	
Circulation areas (including lobbies and atria)	110 lux at floor level. ⁹
Storage spaces	
Dining areas	110 lux at task surface. ⁹
Lounges	
Restrooms	

For Dwelling Units:

Option 1: Promote Visual Acuity

The following requirements are met:

- a. Lighting is installed in kitchens and bathrooms to comply with the illuminance thresholds specified in one of the following lighting reference guidelines:
 1. IES Lighting Library, Lighting Applications Standards Collection.³
 2. ISO 8995-1:2002(E) (CIE S 008/E:2001).⁶
 3. GB50034-2013.⁷
 4. CIBSE SLL Code for Lighting.⁸

b. For spaces where lighting is not installed, the following is provided to all tenants:

1. Illuminance thresholds for common tasks conducted in spaces
2. Specifications, quantity and location of light fixtures required to meet light levels based on sample layout

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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L03 CIRCADIAN LIGHTING DESIGN | O (MAX : 3 PT)

Intent : Support circadian and psychological health through indoor daylight exposure and outdoor views.

Summary : This WELL feature requires projects to provide users with appropriate exposure to light for maintaining circadian health and aligning the circadian rhythm with the day-night cycle.

Issue :

Our body's circadian rhythms are kept in sync by various cues, including light. Humans have evolved to base their circadian rhythms around the natural light-dark patterns associated with daytime and night-time. However, with humans being indoors for extended periods of time, exposure to adequate levels of light have been compromised as typical indoor electric light levels often do not equate to the amount of light the human body traditionally receives outdoors.¹ Light deficiencies affects the functioning of the circadian system and quality of sleep. Disruption of circadian rhythm has been linked with obesity, diabetes, depression and metabolic disorders.²⁻⁴ Exposure to light at night has also been associated with negative health effects, such as breast cancer, circadian phase disruption and sleep disorders.^{5,6}

Solutions : Since circadian response of humans to light is dependent on the light that enters the eye, factors such as spectral properties of the light, brightness levels, duration, and timing of exposure should be considered. The light levels must be achieved on the vertical plane, at the eye level of the occupant to simulate the light entering the eye of the user.^{7,8} It is also important to consider the duration of exposure to light, as well as the timing of exposure. Stimulating the circadian system at night through exposure to bright light can negatively impact sleep quality.^{9,10}

PART 1 MEET LIGHTING FOR DAY-ACTIVE PEOPLE (MAX : 3 PT)

For All Spaces except Dwelling Units & Guest Rooms:

For workstations used during the daytime, electric lighting is used to achieve the following thresholds:

- a. The following light levels are achieved for at least four hours (beginning by noon at the latest) at a height of 45 cm above the work-plane for all workstations in regularly occupied spaces:

Tier	Threshold	Threshold for Projects with Enhanced Daylight	Points
1	At least 150 EML [136 melanopic EDI]	OR At least 120 EML [109 melanopic EDI] and either L05 Part 1 or L06 Part 1	1
2	At least 275 EML [250 melanopic EDI] ¹¹	OR At least 180 EML [163 melanopic EDI] and either L05 Part 1 or L06 Part 1	3

- b. The light levels are achieved on the vertical plane at eye level to simulate the light entering the eye of the occupant.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

For Dwelling Units & Guest Rooms:

The following requirements are met in each dwelling unit:

- a.

Electric lighting is used to achieve the following light levels:

Tier	Threshold	Threshold for Projects with Enhanced Daylight	Points
1	At least 150 EML [136 melanopic EDI]	OR At least 120 EML [109 melanopic EDI] and either L05 Part 1 or L06 Part 1	1
2	At least 275 EML [250 melanopic EDI] ¹¹	OR At least 180 EML [136 melanopic EDI] and either L05 Part 1 or L06 Part 1	3

- b. The light levels are dimmable. If automated lighting is used, it is automatically dimmed after 8:00 pm.
- c. The light levels are achieved in living rooms and kitchens at a height of 140 cm in the center of the room. For studio apartments and guestrooms without living rooms, also test in the center of the room. If workstations are present, light levels are achieved at a height of 45 cm above the work-plane.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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L04 ELECTRIC LIGHT GLARE CONTROL | O (MAX : 2 PT)

Intent : Minimize glare caused by electric light.

Summary : This WELL feature requires projects to manage glare by using strategies, such as calculation of glare and choosing the appropriate light fixtures for the space.

Issue : Glare is defined as excessive brightness of the light-source, excessive brightness-contrasts and excessive quantity of light^{1,2} and is an integral part of lighting design. Reducing glare improves the visual experience of the occupants in the space. Glare has been associated with a host of health issues that range from visual discomfort and eye fatigue to headaches and migraines.^{1,3} Studies have also shown that glare can lead to visual impairment and discomfort, which can cause accidents in the workplace. Individuals under the age of 50 are more sensitive to glare.⁴ Since a substantial section of the workforce falls into this age group, it is important to address glare to avoid visual fatigue and glare-induced headaches.

Solutions : Electric lighting, the light source, type of luminaires and lighting layout can help to reduce glare.

PART 1 MANAGE GLARE FROM ELECTRIC LIGHTING (MAX : 2 PT)

For All Spaces except Industrial:

Option 1: Luminaire considerations

All luminaires within regularly occupied spaces (excluding wall wash fixtures, concealed fixtures, emergency lighting and decorative fixtures installed as specified by the manufacturer) meet one of the following requirements when measured at light output representative of regular use conditions:

- a. 100% of light is emitted above the horizontal plane.
- b. Classified with Unified Glare Rating (UGR) of 19 or lower.
- c. Luminance that does not exceed 6,000 cd/m² at any angle between 45 and 90 degrees from nadir.

OR

Option 2: Space considerations

All regularly occupied spaces meet the following requirement:

- a. Classified with Unified Glare Rating (UGR) of 19 or lower.

For Industrial:

Option 1: Luminaire considerations

All luminaires within regularly occupied spaces (excluding wall wash fixtures, concealed fixtures, emergency lighting and decorative fixtures installed as specified by the manufacturer) meet one of the following requirements when measured at light output representative of regular use conditions:

- a. 100% of light is emitted above the horizontal plane.
- b. Classified with Unified Glare Rating (UGR) of 19 or lower.
- c. Luminance that does not exceed 6,000 cd/m² at any angle between 45 and 90 degrees from nadir.

OR

Option 2: Space considerations

All regularly occupied spaces meet the following requirement:

- a. Unified Glare Rating (UGR) of 19 or lower.

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L05 DAYLIGHT DESIGN STRATEGIES | O (MAX : 4 PT)

Intent : Provide daylight exposure indoors through design strategies.

Summary : This WELL feature requires projects to design spaces to integrate daylight into indoor environments, so that daylight may be used for visual tasks along with electric lighting. It also provides individuals with a connection to outdoor spaces through windows.

Issue : Humans spend approximately 90% of their time indoors.¹ The way buildings are designed has an enormous impact on occupant exposure to daylight.² Exposure to daylight has been proven to have a substantial impact on mood, circadian health and productivity.^{2,3} Studies have associated lack of exposure to daylight with a disruption in the circadian rhythms of humans and a decrease in quality of sleep.² Rooms with large windows also reduce recovery time for patients suffering from severe depression and those recuperating after heart attacks, compared to similarly afflicted patients in rooms with windows facing buildings or other obstructions.⁴⁻⁶ Window characteristics have also been found to have an impact on students' performance in schools.⁷ Students have been shown to perform better, when they have access to windows that can be opened.⁸ Studies also show strong links between quality of views in office environments and performance of employees.⁹ Indoor spaces with daylight exposure have been shown to have fewer bacteria, compared to spaces without any light exposure.¹⁰

Solutions : Building design and interior layout have a substantial impact on the amount of daylight in an indoor space. Indoor daylight access should be accounted for at all stages of building planning from architectural and façade design to interior design and layout. Indoor daylight planning should be coupled with glare control strategies, such as integrating shading for all windows.

PART 1 IMPLEMENT DAYLIGHT PLAN (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The following requirement is met:

- a. The project demonstrates that the following conditions are achieved:

Tier	Interior Layout	Façade Design	Points
1	70% of all workstations are within 7.5 m of envelope glazing. Visible light transmittance (VLT) is greater than 40%.	OR Envelope glazing is no less than 15% of the regularly occupied floor area. Visible light transmittance (VLT) of windows is greater than 40%.	1
2	70% of all workstations are within 5 m of envelope glazing. Visible light transmittance (VLT) is greater than 40%.	OR Envelope glazing is no less than 25% of the regularly occupied floor area. Visible light transmittance (VLT) of windows is greater than 40%.	2

For Dwelling Units & Guest Rooms:

The following requirement is met:

- a. One of the following requirements is met in each dwelling unit:

Tier	Vertical Envelope Glazing Requirements	Points
1	Vertical envelope glazing is no less than 15% of the regularly occupied floor area of each dwelling unit. Visible light transmittance (VLT) is greater than 40%	1
2	Vertical envelope glazing is no less than 25% of the regularly occupied floor area of each dwelling unit. Visible light transmittance (VLT) is greater than 40%	2

PART 2 INTEGRATE SOLAR SHADING (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The following requirements are met in regularly occupied spaces:

- a. All vertical envelope glazing has shading that meets one of the following:

Tier	Type of Shading	Points
1	Manual shading controllable by regular occupants at all times. Shades are regularly opened once a day for all days that the project is in use	1
2	Shading is automated to prevent glare	2

For Dwelling Units & Guest Rooms:

The following requirements are met in dwelling units:

- a. All vertical envelope glazing has shading that meets one of the following:

Tier	Type of Shading	Points
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1	Manual shading is controllable by regular occupants at all times.	1
2	Shading is automated to prevent glare.	2

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L06 DAYLIGHT SIMULATION | O (MAX : 2 PT)

Intent : Ensure indoor daylight exposure through daylight simulation strategies.

Summary : This WELL feature requires projects to conduct daylight simulation calculations to make informed decisions around fenestration and shading, so as to provide appropriate daylight exposure for occupants.

Issue : The way buildings are designed has an enormous impact on occupant exposure to daylight.¹ Exposure to daylight has been proven to have a substantial impact on mood, circadian health and productivity.^{1,2} Studies have associated lack of exposure to daylight with a disruption in the circadian rhythms of humans and a decrease in sleep quality. Students have been shown to perform better when they have access to windows that can be opened.³ Indoor spaces with daylight exposure have been shown to have fewer bacteria, compared to spaces without any light exposure.⁴

Solutions : Building design and planning has a substantial impact on the amount of daylight in an indoor space. With the myriad of solutions available for daylight ingress and shading, indoor daylight exposure can be optimized by conducting simulations. Daylight simulations can inform choices for shading, support fenestration design, as well as the interior layout of a spaces.⁵⁻⁷

PART 1 CONDUCT DAYLIGHT SIMULATION (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The project demonstrates, through computer simulations, that the following conditions are achieved:

a.

The total floor area of regularly occupied spaces achieves one of the following targets:

Tier	Calculations per IES LM-83	Calculations per Annex A of CEN 17037	Points
1	sDA _{300,50%} > 55%	OR Target illuminance of 300 lux is achieved for >50% of regularly occupied areas throughout 50% of daylit hours of the year	1
2	sDA _{300,50%} > 75%	OR Target illuminance of 300 lux is achieved for >50% of total area and average illuminance 100 lux is achieved for >95% of total area throughout 50% of daylit hours of the year	2

For Dwelling Units & Guest Rooms:

The project demonstrates, through computer simulations, that the following conditions are achieved:

a. The total floor area of regularly occupied spaces of each dwelling unit achieves one of the following targets:

Tier	Calculations per IES LM-83	Calculations per Annex A of CEN 17037	Points
1	sDA _{300,50%} > 55%	OR Target illuminance of 300 lux is achieved for >50% of individual unit area throughout 50% of daylit hours of the year	1
2	sDA _{300,50%} > 75%	OR Target illuminance of 300 lux for >50% of individual unit area and average illuminance 100 lux is achieved for >95% of individual unit area throughout 50% of daylit hours of the year	2

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L07 VISUAL BALANCE | O (MAX : 1 PT)

Intent : Create lighting environments that enhance visual comfort.

Summary : This WELL feature requires projects to develop and implement strategies to create a visually comfortable lighting environment.

Issue : A lighting environment may utilize different kinds of light sources, including daylighting and different types of electric lighting technology. Lighting is planned across spaces to achieve the required ambience and energy efficiency goals. However, fluctuating light levels impact the visual comfort of occupants and could lead to eye fatigue.¹ Sudden increases and decreases in brightness can cause a high level of visual discomfort.² Constant changes in lighting have been linked to distraction from the task at hand, which can decrease productivity and impact well-being.³

Solutions : Development of a lighting layout and operations schedule to complement the lighting design in a space is key to increasing the comfort of users. Evidence suggests that thoughtful planning of lighting in a space that takes into account color temperature, daylight and electric light supports a visually comfortable lighting environment.^{4,5} Consideration of the ages of users, tasks performed and existing physical features in the space are also integral to creating a productive space.

PART 1 BALANCE VISUAL LIGHTING (MAX : 1 PT)

For All Spaces:

Option 1: Parameters for visual balance

Ambient lighting in all regularly occupied spaces meets at least three of the following requirements:

- a. Horizontal and vertical luminance contrast ratios for an ambient light system is no more than 10:1 between adjacent independently controlled zones.
- b. Illuminance uniformity ratio of at least 0.4 or 1:2.5 (minimum light level: average light level) is achieved on any horizontal task plane within a space.
- c. One of the following:
 1. A lighting automation system is in use and automatic changes in lighting characteristics, such as light levels, changes in color and distribution take place over a period of at least 10 minutes.
 2. A lighting automation system is not in use.
- d. The Correlated Color Temperature (CCT) in each room for similar fixtures is consistent (± 200 K) at any point of time.

OR

Option 2: Design for visual balance

Lighting is designed by a lighting professional and takes into account the following considerations:

- a. Luminance ratios on vertical and horizontal adjacent zones.
- b. Illuminance uniformity on horizontal task planes.
- c. Changes in lighting characteristics, such as light levels, changes in color and distribution.
- d. Correlated Color temperature (CCT) of lights used.

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L08 ELECTRIC LIGHT QUALITY | O (MAX : 3 PT)

Intent : Enhance visual comfort and minimize flicker for electric light.

Summary : This WELL feature requires projects to take into account characteristics of electric light used in the space, such as color rendering and flicker.

Issue : Humans have evolved to depend on the sun as the main and ideal source of light. Humans are tuned to the color rendering provided by daylight and recognize colors in association with daylight.¹ Color can impact peoples' cognition and behavior.² Using electric light with high color rendering can improve people's perception of a space, and low color rendering can impact the ability to differentiate between objects and perceive the surroundings accurately. Electric lighting used indoors also has low frequencies of flicker that are not present in daylight. Flicker has been associated with eye strain, headaches, migraines and epileptic seizures.³⁻⁶ In 2016, migraines accounted for 16 million disability adjusted life years (DALYs) in men and 30 million DALYs in women.⁷

Solutions : Identifying and utilizing lighting fixtures that emit a high quality of light and do not display signs of flicker contributes to a comfortable and healthy space. Light fixtures with higher color rendering emit light that shows colors realistically. The CIE General Color Rendering Index ("CRI") and measures from IES TM-30 are commonly used to communicate the color rendering properties of a light source.

PART 1 ENHANCE COLOR RENDERING QUALITY (MAX : 1 PT)

For All Spaces except Circulation Areas:

All luminaires in occupiable spaces (except decorative fixtures, emergency lights and other lighting for signage) meet at least one of the following color rendering requirements. If tunable white lighting is used, the requirements are met at 1000 K intervals from the lower end (with a minimum of 2700 K) to the higher end (with a maximum of 5000 K):

- a. CRI (R_a) ≥ 90 .
- b. CRI (R_a) ≥ 80 and R9 (R_9) ≥ 50 .
- c. IES TM-30 P1 (i.e., $R_f \geq 78$, $R_g \geq 95$, $-1\% \leq R_{cs,h1} \leq 15\%$).¹⁰

For Circulation Areas:

All luminaires (except decorative fixtures, emergency lights and other special-purpose lighting) meet at least one of the following color rendering requirements:

- a. CRI (R_a) ≥ 80 .
- b. P2 (i.e., $R_f \geq 75$, $R_g \geq 92$, $-7\% \leq R_{cs,h1} \leq 19\%$).

PART 2 MANAGE FLICKER (MAX : 2 PT)

For All Spaces:

All luminaires (except decorative lights, emergency lights and other lighting for signage) and their controls within occupiable spaces meet at least one of the following requirements:

- a. Classified as "reduced flicker operation" per California Title 24, when tested according to the requirements in Joint Appendix JA-10.⁸
- b. Recommended practices 1, 2 or 3 as defined by IEEE standard 1789-2015 LED.⁹
- c. Pst LM ≤ 1.0 and SVM ≤ 0.6 .

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L09 OCCUPANT LIGHTING CONTROL | O (MAX : 3 PT)

Intent : Provide individuals with access to customizable lighting environments.

Summary : This WELL feature requires projects to implement innovative lighting strategies that take into account personal preferences of users, as well as their interaction with the physical space.

Issue : Humans spend about 90% of their time indoors and the ambience of the indoor environments has an impact on the well-being and productivity of occupants. Lighting in a space has been shown to have a positive impact on mood and cognitive performance.^{1,2} Lighting environments that are customizable by individuals have been shown to improve satisfaction levels.³ The ages of occupants have an impact on the light levels required for visual acuity.^{4,5} Lighting guidelines recommend twice the light levels for individuals over the age of 65 years, compared to individuals between the ages of 25 years and 65 years.^{5,6} As humans age, the transmission of light through their lens is reduced. This is due to age related changes, including increased light absorption by the lenses smaller pupil size, increased scattering of light due to thicker lenses and yellowing of the lenses.^{7,8}

Solutions : Developing a lighting environment that not only seeks to satisfy the visual and circadian requirements of individuals, but also creates a customizable environment helps to improve productivity, mood and well-being. Innovative lighting strategies, including the customization of a user's immediate environment, can contribute to occupant satisfaction with the space. Creating zones with lighting conditions that are distinct from the lighting in regular workspaces, can create a comfortable and informal environment that individuals can utilize for social interaction. Lighting environments can help to improve mental health, reduce stress and improve visual acuity.^{1,9}

PART 1 ENHANCE OCCUPANT CONTROLLABILITY (MAX : 2 PT)

For All Spaces:

1: Lighting zones

Ambient lighting systems meet the following requirement:

a.

All regularly occupied spaces contain lighting zones as shown in the table below (note: individual rooms smaller than the areas below and/or that have occupancies less than those listed in the table are considered separate zones):

Tier	Number of Zones		Number of Zones	Points
1	One per 60 m ²	OR	One per 10 occupants	1
2	One per 30 m ²	OR	One per 5 occupants	2

2: Lighting control system

Each lighting zone meets the following requirements:

- Lighting systems have at least three lighting levels or scenes that allow for changes in light levels.
- Lighting systems have the ability to change at least one of the following:
 - Color.
 - Correlated color temperature (CCT).
 - Distribution of light by controlling different groups of lights or through preset scenes.
- All regular occupants have control over their immediate lighting environment through at least one of the following:
 - Manual controls (e.g. switches or control panels) located in the same space as each lighting zone.
 - Digital interface available on a computer or phone.
- Lighting for presentation or projection walls are separately controlled.

Note :

Projects may identify spaces within the project boundary where lighting controls are limited to certain regular occupants (e.g., facilities staff, office manager) provided that these spaces do not make up more than 10% of the project size.

PART 2 PROVIDE SUPPLEMENTAL LIGHTING (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

1: Supplemental lighting availability

The project meets the following requirements:

- Supplemental light fixtures (e.g., task lights) are provided upon request to all employees at no cost and requests are fulfilled within eight weeks.
- At least one supplemental light fixture is available for trial purposes.

2: Supplemental lighting requirements

Supplemental light fixtures meet the following requirements:

- Light levels are controllable by occupants, independently from the ambient lighting system.

- b. The location of the light is adjustable by users of the workstation.
- c. Under intended use, the light-emitting element is not visible to users.

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APPENDIX L1:

The impact of light on the circadian system can be measured using a metric called Equivalent Melanopic Lux. This metric was proposed by Lucas and others (Lucas et al., "Measuring and using light in the melanopsin age." Trends in Neuroscience, Jan 2014). The authors provided a toolbox which for a desired spectrum derives equivalent "α-opic" lux for each of the five photoreceptors in the eye (three cones, rods, and the ipRGCs).

EML is dependent on the light intensity as well as the spectral power density of light at the measurement point. Given a spectrum of light, each equivalent α-opic lux is related to each other by a constant called Melanopic Ratio (R).

To calculate the equivalent melanopic lux (EML), multiply the photopic lux (L) designed for or measured in a building by this constant (R): $EML = L \times R$.

Projects may use spreadsheets published by the [authors of the journal article](#) or [IWBI](#) for this calculation.